

The table structure is as follows:

-- Students Table

```
CREATE TABLE students (  
  student_id INT PRIMARY KEY,  
  student_name VARCHAR(50),  
  gender VARCHAR(10),  
  city VARCHAR(50),  
  join_year INT );
```

-- Courses Table

```
CREATE TABLE courses (  
  course_id INT PRIMARY KEY,  
  course_name VARCHAR(100),  
  department VARCHAR(50) );
```

-- Marks Table

```
CREATE TABLE marks (  
  mark_id INT PRIMARY KEY,  
  student_id INT,  
  course_id INT,  
  marks INT,  
  FOREIGN KEY (student_id) REFERENCES students(student_id),  
  FOREIGN KEY (course_id) REFERENCES courses(course_id) );
```

-- Insert into

```
students INSERT INTO students (  
student_id, student_name,  
gender, city, join_year) VALUES (1, 'Anu', 'F', 'Tumakuru', 2024),  
(2, 'Ravi', 'M', 'Bengaluru', 2023),  
(3, 'Kiran', 'M', 'Tumakuru', 2024),  
(4, 'Sneha', 'F', 'Mysuru', 2023),  
(5, 'Manu', 'M', 'Tumakuru', 2022);
```

-- Insert into

```
courses INSERT INTO courses (  
course_id, course_name, department) VALUES  
(101, 'SQL Basics', 'Computer Science'),  
(102, 'Excel for Analysts', 'Commerce'),  
(103, 'Statistics', 'Mathematics');
```

-- Insert into

```
marks INSERT INTO marks (  
mark_id, student_id, course_id, marks) VALUES
```

(1, 1, 101, 85),
(2, 2, 101, 72),
(3, 3, 101, 90),
(4, 4, 102, 88),
(5, 5, 103, 67),
(6, 1, 103, 79),
(7, 2, 102, 81);

1. Display all students.

Select * from students;

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	student_id	student_name	gender	city	join_year
▶	1	Anu	F	Tumakuru	2024
	2	Ravi	M	Bengaluru	2023
	3	Kiran	M	Tumakuru	2024
	4	Sneha	F	Mysuru	2023
	5	Manu	M	Tumakuru	2022
✱	NULL	NULL	NULL	NULL	NULL

students 1 ×

2. Display only student_name and city from students table.

select student_name,city from students;

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	student_name	city			
▶	Anu	Tumakuru			
	Ravi	Bengaluru			
	Kiran	Tumakuru			
	Sneha	Mysuru			
	Manu	Tumakuru			

students 2 ×

2. Show all courses.

select * from courses;

Result Grid				Edit:	Export/Import:	Wrap Cell Content:
	course_id	course_name	department			
▶	101	SQL Basics	Computer Science			
	102	Excel for Analysts	Commerce			
	103	Statistics	Mathematics			
*	NULL	NULL	NULL			

courses 3 ×

4. Display students who are from Tumakuru.

```
select * from students where city='Tumakuru';
```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
student_id	student_name	gender	city	join_year
1	Anu	F	Tumakuru	2024
3	Kiran	M	Tumakuru	2024
5	Manu	M	Tumakuru	2022
* NULL	NULL	NULL	NULL	NULL

students 4 x

5. Display students who joined in 2024.

```
select * from students where join_year = '2024';
```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
student_id	student_name	gender	city	join_year
1	Anu	F	Tumakuru	2024
3	Kiran	M	Tumakuru	2024
* NULL	NULL	NULL	NULL	NULL

Level 2: Filtering

6. Show students whose gender is F.

```
select * from students where gender = 'f';
```

Result Grid

Filter Rows:

Edit:

Export/Import:

	student_id	student_name	gender	city	join_year
▶	1	Anu	F	Tumakuru	2024
	4	Sneha	F	Mysuru	2023
*	NULL	NULL	NULL	NULL	NULL

7. Show marks greater than 80.

```
select * from marks where marks > 80;
```

Result Grid		 Filter Rows:	Edit: 		
	mark_id	student_id	course_id	marks	
	1	1	101	85	
	3	3	101	90	
	4	4	102	88	
	7	2	102	81	
	NULL	NULL	NULL	NULL	

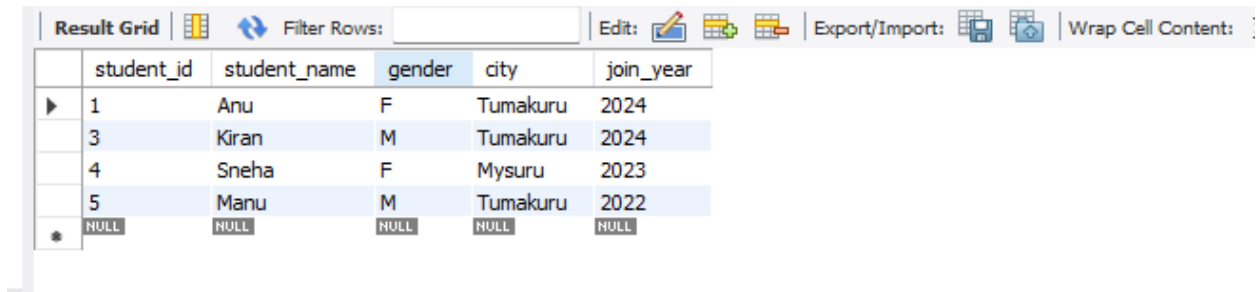
8. Display course names from Commerce department.

```
select course_name from courses where department = 'commerce';
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
course_name			
Excel for Analysts			

9. Show students who are not from Bengaluru.

```
select * from students where city != 'bengaluru';
```

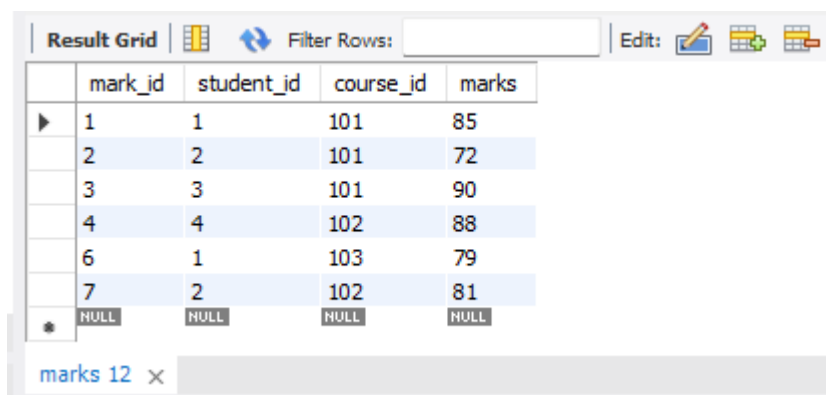


The screenshot shows a database query result grid with the following columns: student_id, student_name, gender, city, and join_year. The results are as follows:

	student_id	student_name	gender	city	join_year
▶	1	Anu	F	Tumakuru	2024
	3	Kiran	M	Tumakuru	2024
	4	Sneha	F	Mysuru	2023
	5	Manu	M	Tumakuru	2022
*	NULL	NULL	NULL	NULL	NULL

10. Display marks between 70 and 90.

```
select * from marks where marks between 70 and 90;
```



The screenshot shows a database query result grid with the following columns: mark_id, student_id, course_id, and marks. The results are as follows:

	mark_id	student_id	course_id	marks
▶	1	1	101	85
	2	2	101	72
	3	3	101	90
	4	4	102	88
	6	1	103	79
	7	2	102	81
*	NULL	NULL	NULL	NULL

marks 12 x

Level 3: Sorting

11. Display all students ordered by student_name ascending.

```
select * from students order by student_name asc;
```

Result Grid					
Filter Rows:					
Edit:   					
Export/Import:  					
Wrap Cell Content: ☐					
	student_id	student_name	gender	city	join_year
▶	1	Anu	F	Tumakuru	2024
	3	Kiran	M	Tumakuru	2024
	5	Manu	M	Tumakuru	2022
	2	Ravi	M	Bengaluru	2023
	4	Sneha	F	Mysuru	2023
*	NULL	NULL	NULL	NULL	NULL

12. Show marks ordered from highest to lowest.

```
select * from marks order by marks DESC;
```

Result Grid				
Filter Rows:				
Edit:   				
Export/Import: 				
	mark_id	student_id	course_id	marks
▶	3	3	101	90
	4	4	102	88
	1	1	101	85
	7	2	102	81
	6	1	103	79
	2	2	101	72
	5	5	103	67
	NULL	NULL	NULL	NULL

marks 15 ×

13. Display students ordered by join_year descending.

```
select * from students order by join_year desc;
```

Result Grid					
Filter Rows:					
Edit:   					
Export/Import:  					
	student_id	student_name	gender	city	join_year
▶	1	Anu	F	Tumakuru	2024
	3	Kiran	M	Tumakuru	2024
	2	Ravi	M	Bengaluru	2023
	4	Sneha	F	Mysuru	2023
	5	Manu	M	Tumakuru	2022
*	NULL	NULL	NULL	NULL	NULL

students 16 ×

Level 4: Aggregate Basics

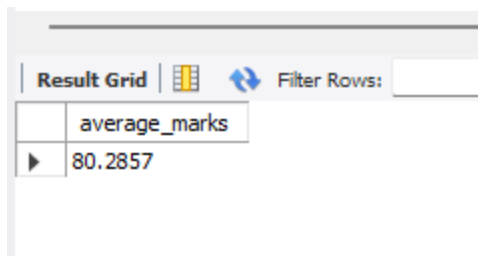
14. Find total number of students.

```
select count(*) AS total_students  
from students;
```

Result Grid	
Filter Rows:	
Export:  Wrap Cell Content: 	
	total_students
▶	5

15. Find average marks.

```
select avg(marks) as average_marks  
from marks;
```

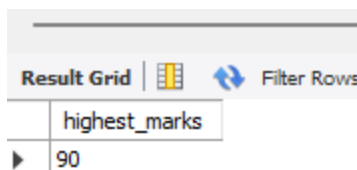


The screenshot shows a database interface with a 'Result Grid' tab. Below the tab, there is a table with one column named 'average_marks' and one row containing the value '80.2857'. To the right of the table, there is a 'Filter Rows' button and a text input field.

average_marks
80.2857

16. Find highest marks.

```
select max(marks) AS highest_marks  
from marks;
```



The screenshot shows a database interface with a 'Result Grid' tab. Below the tab, there is a table with one column named 'highest_marks' and one row containing the value '90'. To the right of the table, there is a 'Filter Rows' button and a text input field.

highest_marks
90

17. Find lowest marks.

```
select min(marks) as lowest_marks
```

from marks;



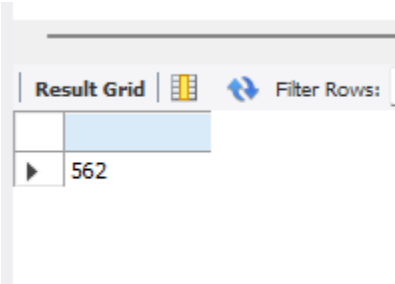
The screenshot shows a SQL query result grid. At the top, there is a header bar with the text "Result Grid" and a "Filter Rows:" button. Below the header, the first column is labeled "lowest_marks". The first row of data contains the value "67".

	lowest_marks
▶	67

18. Find total marks scored by all students.

select sum(marks) AS total_marks

from marks;



The screenshot shows a SQL query result grid. At the top, there is a header bar with the text "Result Grid" and a "Filter Rows:" button. Below the header, the first column is labeled "total_marks". The first row of data contains the value "562".

	total_marks
▶	562